ODYX Products

❖ ODYX P1-26 printer Specifications

- Print Screen: 6.8” 9k LCD Ultra-Clear Monochrome Screen

  Benefit:

  - Very High Clarity So More Precise Details
  - Improves Marginal Accuracy

- X-Y Axis Accuracy: 18 µm

  Benefit:

  - Very High Horizontal Precision

    Better Detail In:

    - Margins
    - Anatomy
    - Suitable For Precise Dental Work

- Resolution: 8520 × 4320 px

  Benefit:

  - Higher Pixel Count So Finer Details
  - Smoother Surface Finish

- Motion System (Z-Axis): Dual Linear Guide Rails + T-Shaped Screw

  Benefit:

  - Higher Stability During Printing
  - Reduces Vibration So Better Accuracy

- Printing Speed: Max Print Speed: 60 mm/h

  Benefit:

  - Faster Production
  - Suitable For Multiple Cases Per Day

- Layer Thickness: 0.01 – 0.1 mm

  Benefit:

  - Highest Precision (Slower)
  - Lower Precision (Faster)
  - Adjustable Depending on the Case

- Screen Lifetime: 2000 Hours

  Benefit:

  - Long Operating Time Before Replacement
  - Lower Maintenance Cost

- Consumable Types

  - Dental Model Resin

- Standard Resin
- High Fidelity Model Resin
- High Hardness
- High Toughness
- Biocompatible Resin
- Castable Wax Resin

Benefit:

- Can Be Used For:
    - Models
    - Splints
    - Surgical Guides
    - Temporaries
    - Castable Applications

- Touch Screen: 5-Inch Capacitive Color Touch Screen

  Benefit:

    - Easy Operation
    - Direct Control of The Printer

- Connectivity: USB / WIFI

  Benefit:

    - Easy And Fast File Transfer

    - Cloud printing supported: Print jobs can be sent over wifi
    - No Ethernet
    - No mobile App

- Printing Size: 153 × 77 × 160 mm

  Benefit:

    - Can Print:
        - Multiple Crowns
        - Full Models
        - Suitable For Batch Production

- Machine Size: 221 × 221 × 404 mm

  Benefit:

    - Compact Size
    - Fits Easily in Dental Clinic

- Light source: Third generation integral light source at 405nm

  Benefit:

    - Provides fast, stable, and complete curing for all standard dental resins.

- Light uniformity >90%

  Benefit:

  - Ensures consistent curing throughout the entire model for greater accuracy, strength, and dependable print quality.

- Simple workflow that can be learned quickly.
- Open material choice keeps ongoing resin cost flexible.
- User interface available in 13 languages
- Net weight 6.6 kg:

  Benefit:

  - Compact and lightweight for easy handling, installation, and efficient workspace utilization.

Software compatibility

- Native slicer: ODYX slicer (Windows 7, 8, 10 x64 and macOS).
- Multi printer management is possible through the ODYX slicer.
- Printing parameters are managed through the ODYX slicer.
- Native print file format: CXDLPV4.

Supported materials

- The printer is compatible with a wide range of dental resin

Workflow and ease of use

- Sand blasted print platform for strong build adhesion.

- High speed release film to lower peel resistance and improve printsuc cess rate.
- Resin vat with a builtin liquid level marking to prevent overfilling &re sin waste.
- Simple setup, designed for daily chair side use.

Maintenance

- LCD screen rated for 2000 hours of use.
- Firmware updates are done manually using USB.

What is not on this printer

- No automatic resin heating.
- No automatic resin refill.
- No AI camera.
- No print monitoring camera.
- No failure detection.
- No resin level sensor (only resin level markings).
- No tilt release technology.
- No automatic leveling.
- No mobile app.
- No smart resin profiles.
- No automatic firmware updates.
- No heated resin tank.
- No chamber heating.

Technical Summary

- High Precision (18 µm)
- 9k Screen → Strong Detail Quality
- Supports Multiple Dental Materials
- Suitable For Batch Production

➢ **Regarding the hardware devices, modifications have been applied exclusively to the ODYX P1-26 printer.**

- An optional small tank and platform (60 × 60 × 100 mm) designed for smaller print jobs such as crowns, bridges, and veneers.
- A switch from NFEP sheets to ACF sheets.
- Rebranding the slicing software under the name ODYX Box.
- The addition of a dispensing channel to the tank.

❖ HALOT Printer Specifications

➢ XY Resolution: (14*19 µm)

Benefit:

- Determines Printing Accuracy
- Lower Value = Finer Details

- Acceptable For Models and Temporary Restorations

➢ Layer Thickness: 0.01 – 0.2 mm

Benefit:

- 0.01 Mm → Higher Detail, Smoother Surface
- 0.2 Mm → Faster Printing
- Allows Balance Between Speed and Quality

➢ Build Volume: 211.68 × 118.37 × 200 mm

Benefit:

- Determines how many objects you can print at once
- Suitable for multiple dental models or several crowns in one batch

➢ Screen Type: 10.1 16k Monochrome LCD

Benefit:

- Faster Printing
- Longer Lifespan
- More Stable Performance

➢ Screen Resolution: 15120*6230 px

Benefit:

- Affects Detail Precision
- Higher Resolution = Better Marginal Accuracy

- Printing Speed:170 mm/h “at 0.20 mm layer “

Benefit:

- Faster Layer Curing
- Reduces Total Printing Time
- Efficient For Batch Production

- Touch Screen: 3.98-inch

Benefit:

- Easy To Operate
- User-Friendly For Clinic Use

- Honeycomb Matrix Light Source (405 nm)

Benefit:

- Delivers highly uniform UV light distribution for fast, accurate, and consistent curing across the entire build area.

- Exposure Intensity: 6,500 µW/cm² (Fixed)

Benefit:

- Provides stable, optimized exposure for reliable curing, high print quality, and consistent dimensional accuracy.

- 92 Intelligent Exposure Zones

Benefit:

- Activates only the LED zones beneath the printed model, reducing energy consumption, minimizing heat generation, and significantly extending LCD screen lifespan.

- Motion System: Moving Resin Vat & Light Source, Fixed Build Plate

Benefit:

- Reduces model movement during printing, improving stability, minimizing layer shifting, and enhancing print accuracy, especially for large or delicate models.

➢ Dual Lead Screws with Dual Linear Rails (Z-Axis)
Benefit:

- Provides exceptional mechanical stability and smooth vertical movement, reducing wobble and improving surface finish and dimensional precision.

➢ Z-Axis Positioning Accuracy: 0.01 mm
Benefit:

- Enables precise layer positioning for highly detailed prints with excellent dimensional accuracy and fine surface quality.

➢ Leveling-Free Design
Benefit:

- Factory-calibrated build platform eliminates manual leveling, reducing setup time, preventing user errors, and ensuring consistent first-layer adhesion.

➢ Quick-Release Resin Vat (Tool-Free Removal)
Benefit:

- Simplifies resin replacement and cleaning, reducing maintenance time and improving workflow efficiency.

➢ Release Film Life: ~30,000 Layers
Benefit:

- Long-lasting release film reduces replacement frequency, lowering maintenance costs and minimizing downtime.

➢ LCD Screen Life: ~3,000 Hours
Benefit:

- Durable monochrome LCD provides reliable long-term performance while reducing maintenance and replacement costs.

- UV Lamp Life: ~20,000 Hours

  Benefit:

  - Long-life UV light source ensures consistent curing performance with minimal maintenance over years of professional use.

- Connectivity
  - WIFI.
  - USB (USB drive and Type B USB port).
  - Creality Cloud support for parameter management and cloud-based print sending.

- File Formats: STL / OBJ

  Benefit:

  - Compatible with dental CAD software (EXOCAD, 3Shape)

- Printer Size: 344 × 331 × 434 mm

  Benefit:

  - Compact desktop design that fits easily into dental clinics and laboratories.
  - Maximizes workspace efficiency while offering a large build volume.

- Weight: 12.93 kg

  Benefit:

  - Provides a stable and durable platform for consistent, reliable printing performance.

- Price: 45,000 to 55,000 EGP

Benefit:

- Very Affordable Entry Into 3D Printing

AFU (Auto Feed Unit) features

- Automatic resin feed in and feed out.
- Real time resin level and bottle weight monitoring.
- Laser distance sensor for accurate level measurement.
- AFU makes long prints easier and reduces manual resin handling.
- Detects the remaining resin in the bottle.
- Resin heating between 30 and 45degrees Celsius.

RFID (Radio Frequency Identification) recognition of official

resin (Piocreat Resin> ODYX Resin to be) bottles. When an official bottle is used, the recommended print parameters are loaded automatically from the cloud.

Software and file compatibility

- HALOT BOX slicer (Creality native).
- CHITUBOX supported. 3 month CHITUBOX Pro subscription includ edwith the printer.
- Network printing supported over WiFi.

Supported materials

- Open material: Works with third party resins with manual parameter entry.
- Official Creality and PioCreat resins can be loaded automaticallythrou gh RFID.
- Types mentioned officially include standard resin, ABS like resin, low odour rigid resin and jewelry resin.

Safety features

- Enclosed printer body with a lid.
- Proximity switch on the lid. The build plate doesn’t move when the lid is open. A door security reminder is shown on screen.
- Z axis limit switch.
- Filter and respirator included in the box

<u>Is It Suitable for Dental Use?</u>

Yes, But with Limitations

- It is not a dedicated dental 3D printer, but it can be used in dentistry by using Dental Resin and for certain applications

<u>Suitable For</u>

- Study Models
- Orthodontic Models
- Surgical Guides

- Splints / Night Guards
- Temporary Crowns

Not Recommended For:

- Final Restoration
- Advanced Implant Surgical Guide
- Full Arch Prosthetics

❖ ODYX Cure Specifications:

- A professional curing unit widely available in the Egyptian dental market.
- While it is a "Professional" grade machine, its biocompatibility status depends entirely on the resin you are using.

The nine key technologies that distinguish the ODYX Cure are:

1. Fast Curing Time: Generally, completes the process in 1–5 minutes, significantly reducing wait times.
2. Adjustable Light Intensity: Power can be adjusted from 5% to 100% to meet the specific requirements of different resin types.
3. 8 Memory Presets: Includes 8 groups of data to save and quickly select customizable cure settings for various materials.
4. Triple-Wavelength UV Light: Features 365nm, 385nm, and 405nm LEDs that can work independently or together to cure a wide range of professional resins.
5. 360° All-Round Curing: Ensures uniform light distribution from all sides to provide a more consistent cure and less warping of the parts.

6. One-Way Mirror Design: Combines a visible observation window with internal specular reflection to maximize light efficiency while protecting the user.
7. Countdown Tips: An adjustable timer from 0–30 minutes with a digital display for precise monitoring.
8. Power Dial (Dual Voltage): A manual switch allows the machine to meet different voltage requirements (115V or 230V) for global compatibility.
9. Simple Operation: A friendly interface with a one-button starts and high-definition LED touch sensor buttons.

Recommended Parameters: "It may vary by resin type

- Surgical Guides: ~3 min
- Temporary Crowns: ~10 min
- Dentures: ~15 min
- Castable Resins: ~3 min
- Standard Models: ~2 min

Design & Durability

- Chamber Size: The effective curing area is 180mm (Diameter) x 120mm (Height).
- Smart Protection: An intelligent sensor automatically stops the process if the cover is opened, protecting the user's skin and eyes.
- Anti-Corrosion Body: The machine is made of anodized metal, which makes it easy to clean if resin drips on it and resistant to wear.

Technical Specifications

- Curing chamber size: 180 x 120 mm.
- Three wavelengths available and selectable by the user: 365 nm, 385 nm and 405 nm.
- Adjustable light intensity from 5% to 100%.
- Timer range: 1 second to 30 minutes.
- Stores up to 8 saved curing profiles combining wavelength, intensity and time.
- Typical curing time for standard dental resins: 1 to 5 minutes.
- Touch control with LED display.
- Input voltage: 100 to 240 V, 50 to 60 Hz.
- Chamber interior treated for corrosion resistance.
- Positioned by the manufacturer as a top post processing tool for the dental industry.
- Fine control over both light intensity and time.

❖ UW-03 Specification:

1. Description

The Creality UW-03 is a 2-in-1 resin post-processing machine designed to simplify and improve the finishing process of SLA, LCD, and DLP resin 3D prints. It combines washing (to remove uncured resin) and UV curing (to fully harden the resin) into one compact device, reducing manual work and improving print quality.

2. Key Features

- 2-in-1 Washing & Curing

Instead of purchasing two separate machines, the UW-03 performs both cleaning and UV curing in one unit, saving workspace and reducing equipment costs.

- High-Speed Vortex Washing

The upgraded high-speed stirring motor creates a strong vortex that continuously circulates the cleaning liquid around the model. This removes uncured resin even from deep cavities and fine details without damaging delicate surfaces.

- 360 UV Curing

The rotating curing platform exposes every side of the printed model to UV light, ensuring consistent and uniform curing while minimizing soft spots or uneven hardening.

- Adjustable Time Settings

Users can select washing and curing durations between 5 and 30 minutes, allowing optimization for different model sizes and resin types.

- UV Safety Cover

The transparent UV shield blocks harmful ultraviolet radiation. If the cover is lifted during curing, the machine automatically pauses to protect the operator.

3. Main Selling Points

- Dual-function wash and cure system
- Large washing and curing capacity

- Powerful vortex cleaning technology
- Uniform 360° UV curing
- Wash models directly on the build plate
- Adjustable speed and timer
- Beginner-friendly operation
- Affordable price compared to industrial post-processing systems

4. Applications

The UW-03 is suitable for:

- Dental laboratories
- Dental clinics
- Jewelry manufacturing
- Industry
- Engineering models
- Educational hubs
- Universities
- Research laboratories
- Hobbyists
- Resin printing businesses

5. Compatible Resin Types

The machine supports all UV-curable resins because it only performs cleaning and post-curing.

6. Washing & Curing Modes

- Washing Mode

The machine agitates the cleaning liquid using a high-speed stirring fan to remove uncured resin from printed models.

- Curing Mode

The washing container is replaced with the curing platform, where UV LEDs fully cure the resin to improve its mechanical properties and durability.

7. Technical Specifications

Washing Size

<u>240 × 160 × 200 mm</u>

Benefit

- Suitable for medium and relatively large resin models.
- Allows multiple small parts to be cleaned.
- Curing Size

<u>Ø220 × 300 mm</u>

Benefit

  - Reduces the need to split large prints into multiple sections.

Machine Dimensions

<u>325 × 303 × 436 mm</u>

Benefit

- Compact enough for desktop use.

Machine Language

- English
- Chinese

Wash/Cure Time

5–30 minutes

Benefit

- Prevents under-cleaning or over-cleaning.
- Prevents under-curing or over-curing.
- Suitable for different resin types and model complexities.

8. Functions

Wash Models on the Build Plate

The printed model can remain attached to the printer's build plate during washing.

Benefits

- Minimizes handling.
- Protects delicate models.
- Reduces the chance of accidental damage.
- Keeps hands away from uncured resin.

9. Basket Washing

Models that have already been removed from the build plate are placed inside the washing basket.

Benefits

- Keeps small parts secure.
- Prevents components from falling into the tank.
- Ensures even exposure to the cleaning liquid.

360° UV Curing

The curing platform rotates continuously under UV illumination.

Benefits

- Uniform curing.
- Better dimensional accuracy.
- Improved strength.
- Better surface finish.
- Reduced tacky surfaces.

Adjustable Timer

Users can choose the operating duration before starting.

Benefits

- Customized processing.
- Improved efficiency.
- Consistent results.

High-Speed Stirring Motor

Creates continuous circulation of the cleaning liquid.

Benefits

- Removes resin from fine details.
- Cleans internal channels.
- Produces stronger washing performance than static soaking.
- Improves cleaning consistency.

UV Safety Cover with Auto Pause

Opening the UV cover immediately pauses curing.

Benefits

- Prevents direct UV exposure.
- Improves workplace safety.
- Protects users’ eyes and skin.

10. Operation

Installation

- Install the washing container or curing platform.
- Connect the power supply.
- Fill the tank with cleaning liquid if washing.
- Place the model.
- Close the UV cover.
- Select mode.
- Start the process.

Washing Workflow

1. Place the model.
2. Fill the tank with IPA or cleaning solution.
3. Select washing mode.
4. Choose speed.
5. Set washing time.
6. Start cleaning.
7. Remove the model after completion.

Curing Workflow

1. Remove the washing tank.
2. Install the curing platform.
3. Place the cleaned model.
4. Close the UV cover.
5. Select curing mode.
6. Set curing time.
7. Start curing.
8. Remove the cured model.

11. Daily Maintenance

- Filter contaminated alcohol.
- Allow resin residue to settle.
- Reuse filtered IPA.
- Clean the washing tank.
- Remove resin sediment.
- Inspect the stirring fan.
- Replace bearings if necessary.

12. Troubleshooting

Common issues include:

- Machine won’t start → Check the power connection and ensure the UV cover is closed.
- Stirring fan doesn’t rotate → Remove debris or clean resin buildup.
- Unusual noise → Tighten the fan or lubricate the motor bearing.

- White residue on prints → Use IPA with >95% concentration and allow the model to dry before curing.

Maintenance Instructions

- Regularly clean the machine.
- Maintain the stirring fan.
- Replace worn bearings.
- Keep the washing tank free of cured resin buildup
- Emergency Shutdown

❖ ODYX Resin Specification

Key Material Properties & Definitions

1. Hardness (Shore D vs. Shore A)

Defines the material's resistance to permanent indentation.

- Shore D: Used for rigid materials (e.g., Crowns, Splints). High values (90–95) indicate high scratch resistance.
- Shore A: Used for soft/flexible materials (e.g., Gingiva masks). Lower values (30–50) indicate a soft, rubber-like feel.

2. Flexural Modulus (MPa)

Measures Stiffness.

- High Modulus: The material is rigid and resists bending (ideal for long-span bridges).
- Low Modulus: The material is flexible.

3. Flexural Strength (MPa)

The maximum stress the material can handle before breaking while being bent.

- Range: 140–160 MPa is considered very strong, essential for permanent restorations.

4. Tensile Modulus & Strength (MPa)

- Modulus: Resistance to being pulled apart (stiffness under tension).
- Strength: The maximum stress sustained before the material snaps under tension. Crucial for Aligners and Splints.

5. Elongation at Break (%)

Measures Ductility. It indicates how much the material can stretch before snapping.

- Low (5–10%): Brittle (Ceramic-like).
- High (100%+): Highly elastic (Gingiva/Soft Tissue-like).

6. Impact Strength (J/m)

The ability of the material to absorb energy during a sudden blow or drop without fracturing. Essential for Dentures and Night Guards

7. Heat Deflection Temperature (HDT) (°C)

The temperature at which the material begins to soften or deform.

- High HDT (>100°C): Can withstand sterilization or thermoforming.
- Low HDT (~60°C): For standard models only.

8. Water Absorption (%)

The amount of fluid the resin absorbs over time.

- Lower is better: Ensures dimensional stability and prevents staining/foul odors in long-term use.

1. Ceramic Crown Resin

Uses:

- Crowns
- Veneers
- Inlays / Onlays

Features

- High hardness and strength good for chewing
- Good esthetics (ceramic-like behavior)
- Temp resistance
- Low skin irritation
- Impact-resistant

Highlight:

- Designed for both permanent and temporary restorations.
- High flexural strength and hardness for long-lasting performance.
- Excellent wear and fracture resistance.
- Low polymerization shrinkage for an accurate marginal fit.
- Superior esthetics with natural tooth shades (A1, A2, A3, BL1, B1, B2).
- High temperature resistance for improved dimensional stability.

2. Ortho Model 2.0

Uses:

- Study models
- Aligner models
- Working models

Features:

- Low Skin Irritation
- Low Skin Allergenic
- Low Shrinkage
- Temperature Resistance
- High Hardness

Highlights:

- High dimensional accuracy for precise orthodontic models.
- Smooth surface finish with excellent detail reproduction.
- Withstands the heat generated during vacuum thermoforming.
- High hardness reduces wear and extends model lifespan.
- Ideal for producing accurate aligner and thermoforming models

3. Surgical Guide Pro

Uses:

- Implant surgical guides

Features:

  - Low skin irritation
  - Low skin allergenic
  - Low shrinkage
  - High transparency
  - High elongation
  - High impact strength

Highlights:

- High transparency provides excellent visibility during surgery.
- Superior flexibility helps prevent cracking during use.
- Excellent impact resistance for increased durability.
- Steam sterilizable up to 135°C without cracking or deformation.
- Designed for biocompatible medical and dental applications.

4. Temporary Restoration Resin

Uses:

- Temporary crowns
- Temporary bridges

Features:

- Easy polishing
- Comfortable for patients
- Good short-term stability

Highlight:

- Specifically developed for temporary restorations.
- Excellent dimensional stability with low shrinkage.
- High mechanical strength for reliable short-term use.
- Good temperature resistance during function.
- Low water absorption (<1.5%), helping maintain color and dimensional stability.
- Available in multiple tooth shades (A1, A2, A3, BL1, B1, B2).
- Smooth surface finish with high printing accuracy.

5. Crown & Bridge Resin

Uses:

- Crowns
- Bridges
- Denture teeth
- Inlays
- Onlays
- Veneers

Features

- Temperature Resistance
- Low Skin Irritation
- Low Allergenic Potential
- Low Shrinkage
- High Hardness
- High Impact Resistance

Highlights

- Excellent mechanical strength for long-term restorations.
- Low polymerization shrinkage ensures an accurate fit.
- High hardness provides excellent wear resistance.
- Good temperature resistance for improved dimensional stability.
- High impact resistance minimizes the risk of fracture.

Resin Certifications:

Resin	CE	FDA	ISO	MSDS	REACH	RoHS	ISO 10993	ISO 13485
Ceramic Crown Resin	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Crown & Bridge Resin	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ortho Model 2.0 Resin	Yes	yes	Yes	Yes	Yes	Yes	Yes	Yes
Temporary Restoration Resin	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Surgical Guide Resin Pro	No	No	Yes	Yes	Yes	Yes	Yes	Yes

❖ ODYX-S1 Scanner Specifications

➢ Technical Details:
- Accuracy: $\leq 20\ \mu m$
- Scanning Speed: 40 seconds for full arch
- Scan depth: Depth 23 mm

➢ Good for
- Margin Deep
- Implant Scan Bodies
- Gingival Contour

➢ Remote Control
- Finish the whole process without touching the screen which make it fast & clean “infection control “

➢ Real Color Scanning
- Good For Diagnosis

➢ Tip
- Rounded Small tip size 20*17: Can reach easily to corners and hard areas with patient comfortability

➢ Resolution
- High-resolution 3D imaging (HD)

Meaning:

- Good surface detail
- Suitable for prosthodontics

➢ Weight
- 270 Grams

Meaning:

- Light weight
- Comfortable for long procedures

➢ System Type

◦ Open system

Meaning:

- Exports files: STL, OBJ
- Compatible with any lab
- No ecosystem restriction

➢ Software

◦ Include AI margin detection

◦ Includes scanning software ex “AI technology “filter out unnecessary data like tongue and lips during scanning

➢ Clinical Applications

◦ Crowns & bridges

◦ Inlays / Onlays

◦ Veneers

◦ Basic implant planning

◦ Orthodontics